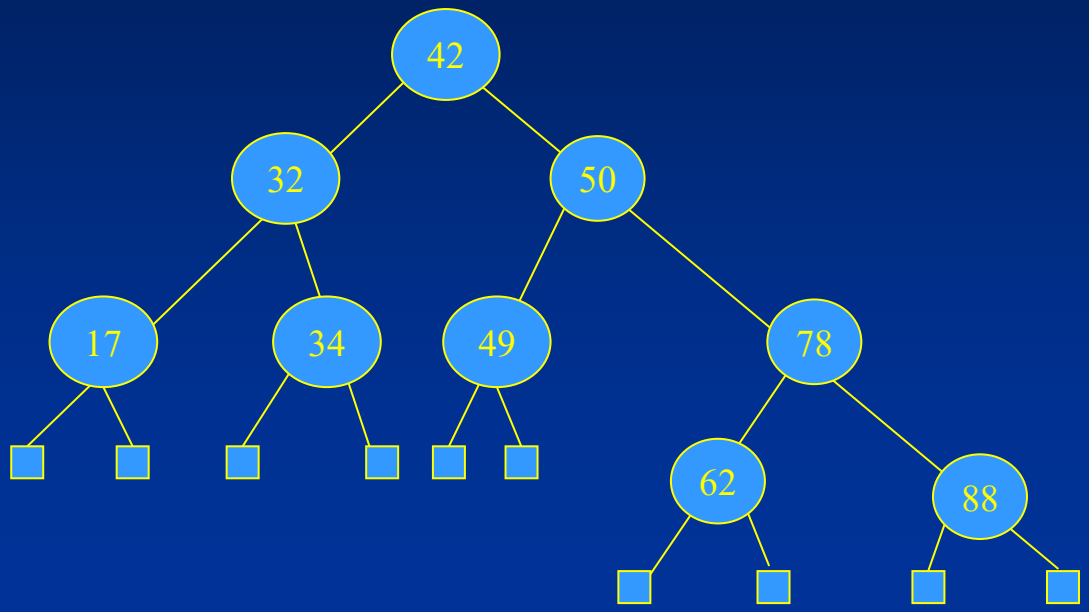
Assignment #7

Balance Trees – Due October 19th before 2:30 and use a tool (not handwritten) for your final answer
No late assignments



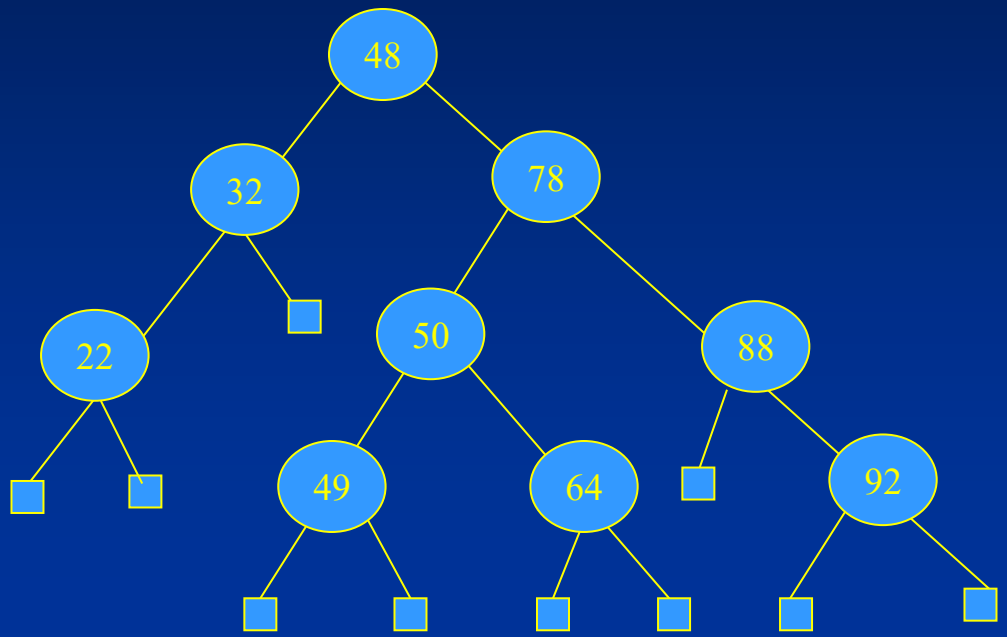
#1 Insertion in this AVL Tree

Insert 64



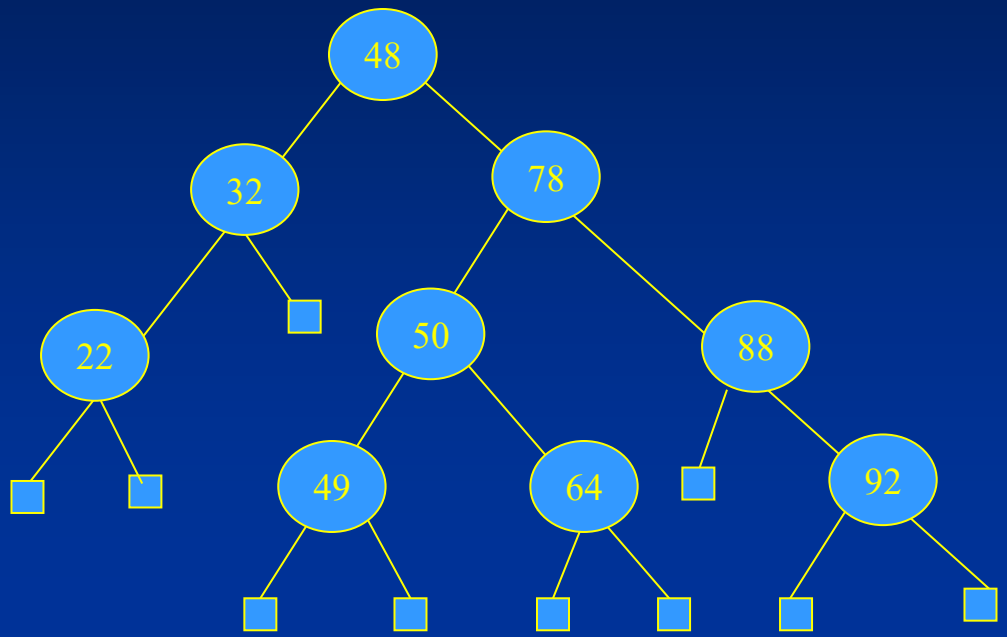
#2 Removal from this AVL Tree

Remove 32



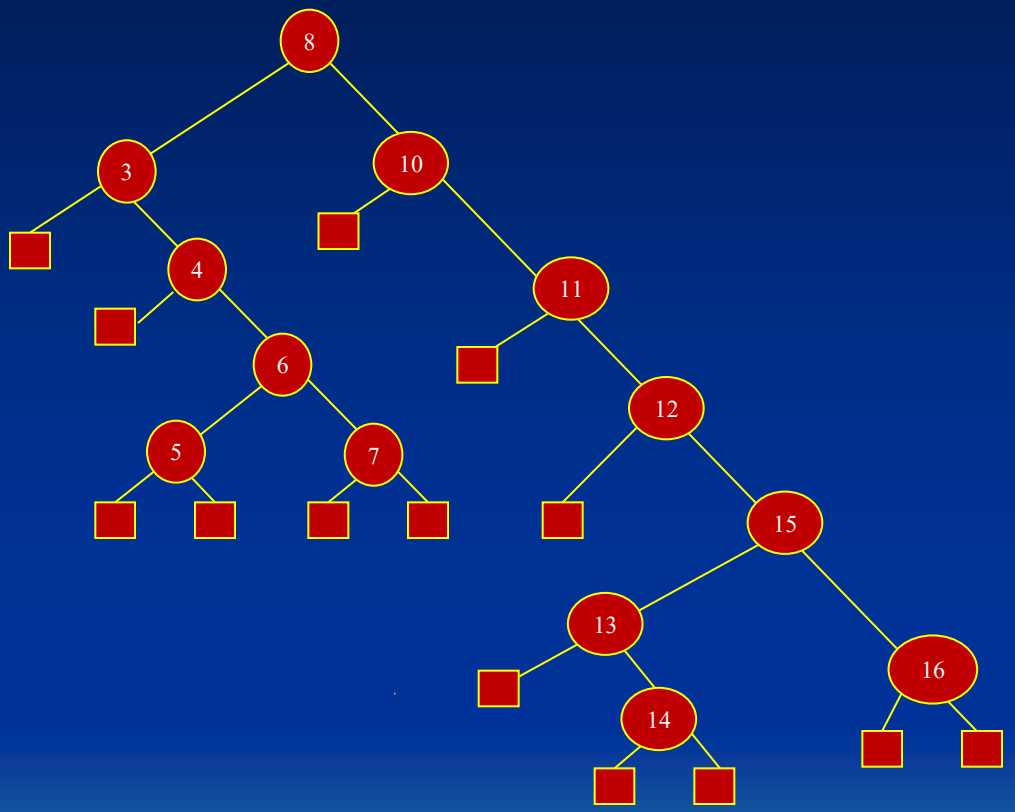
#3 Removal from this AVL Tree

Remove 22



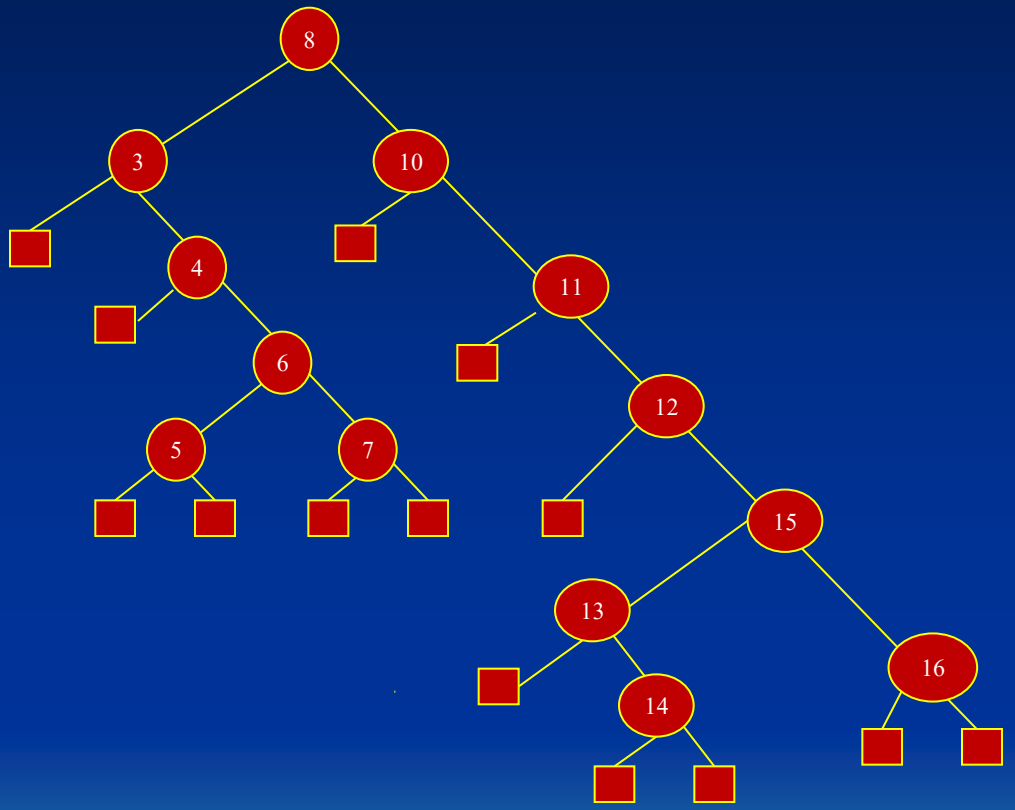
#4 Splaying

Splay 16



#5 Splaying

Delete 7



#6 Insertion into a Splay Tree

Insert 14, 44, 15, 55, 45, 85, 75, 65, 99 into a Splay tree

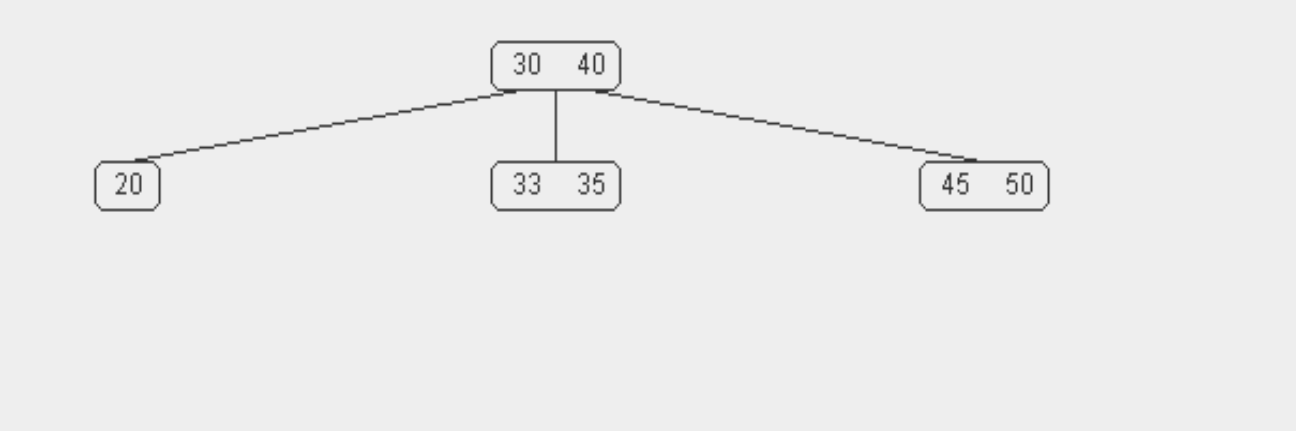


#7 Insertion into a 2-4 Tree

Insert 13, 44, 14, 55, 45, 85, 77, 66, 98
into a 2-4 tree

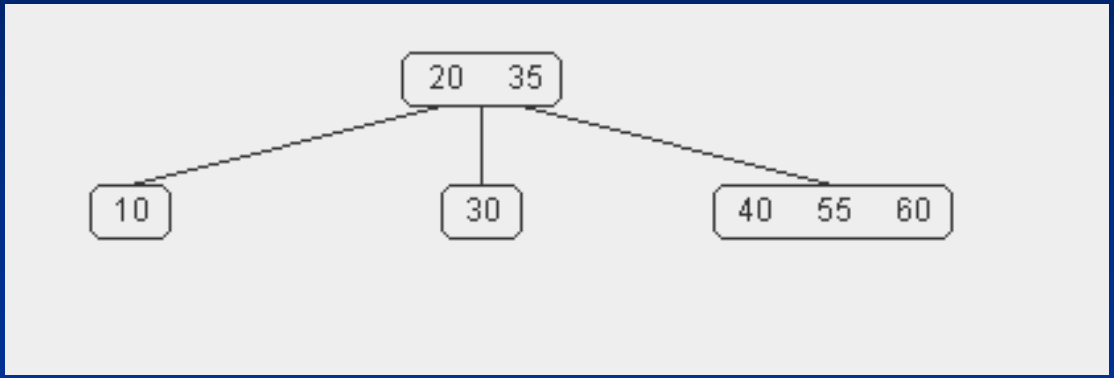


#8 Delete from a 2-4 Tree



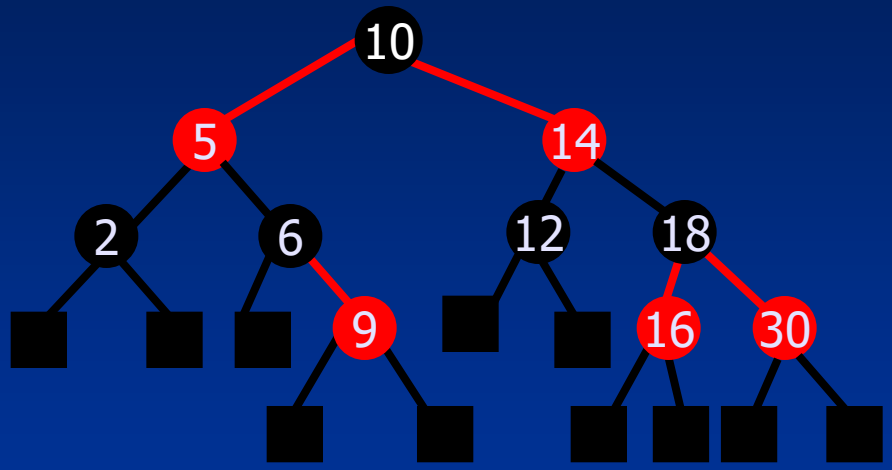
Remove node 50 and then node 45 from this 2-4 tree (show all steps)

#9 Delete from a 2-4 Tree



Remove node 35 from this 2-4 tree

#10 Conversion



Convert this red-black tree to a 2-4 tree